

Welcome to SM00146!



Muhammad Arif
muhammad.arif@gu.se



**Computational
Multi-Omics Lab**
University of Gothenburg



What is this course about

- Systems Biology / Omics Analysis Course
 - For non-computationally-oriented PhD Students/Researchers
 - Don't worry if you can't code, almost everyone is on the same boat! 😊
- **Focus:** How we can use Omics Data and Computational Analysis to answer biological questions
 - Genomics, transcriptomics, metabolomics, proteomics
 - Systems biology
- Combination of theory + hands-on + scientific discussions

**You do not need to become a computer scientist.
You should become able to understand, question, and use
omics analyses in your own research.**

What will not be discussed

- Experimental Parts:

- Library prep
- RNA-extraction
- Randomization of samples, etc etc

(You guys know about this much better than I do)

- I will also not discuss about the quantification of the data, most companies provide this now
 - Drop me an email if you need to do this and I can guide you.
- Software or algorithm developments

What should you be able to do by the end?

You can read the ILO, but here's what I expect

Understand

basic workflows of major omics analyses

Interpret

PCA, volcano plots, heatmaps, enrichment, networks

Recognize

why QC, normalization, metadata and statistics matter

Evaluate

omics analyses in scientific papers

Connect

omics approaches to your own PhD project

Communicate

more effectively with computational collaborators

Know what to ask when you see, generate, or receive omics data.

How the course is organized

Theory

What data represent
Key assumptions
Common methods
What can go wrong

Hands-on

Real datasets
Guided notebooks
Run, inspect,
interpret
Ask why each step
matters

Journal club

Read papers
critically
Discuss omics
design
Opponent questions
Constructive critique

Your research

Which omics layer
helps?
What question
can it answer?
What validation is
needed?

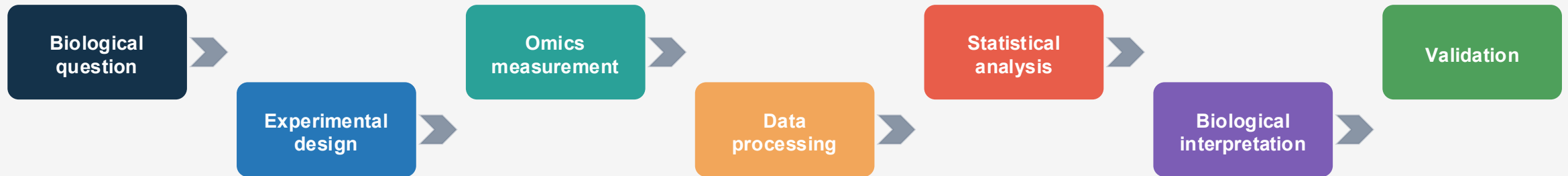
**The lecture is the “why”.
The notebook is the “how”.
The discussion is where you learn to judge the evidence.**

Schedule and Important Dates

Monday	Tuesday	Wednesday	Thursday	Friday
Transcriptomics Lecture 9.15-12.00	Proteo/Metabolomics Lecture 10.00-12.00	Genomics Lecture 9.15-12.00	Metabolomics+ MultiOmics Workshop 9.15-12.00	Group Presentation and closing 09.15-12.00
Transcriptomics Workshop 13.15-15.00	Guest Lectures 13.15-15.00	Genomics Workshop 13.15-15.00	Group Discussion 13.15-15.00	

- **18 September, 23.59: Deadline for written assignment**
- On Monday and Wednesday afternoon, I will stay until 16.00 in case you have any practical/workshop questions
- **Group discussion:** optional to be in class, but I will be available to answer questions

What this course is really about



- A sophisticated analysis cannot rescue a weak biological question.
- Significant does not automatically mean biologically important.
- Omics produces evidence and hypotheses, not automatic truth.
- Interpretation and validation matter as much as computation.

Tools and resources

Course website / GitHub
SysBioPhD.multiomics.se

Posit Cloud
Jupyter notebooks + shared
workspace

Canvas
Announcements +
assignments + papers

- Use the course website as the main reference for notebooks and materials.
- Keep Posit Cloud invitation links private
- Download or save important work from notebooks after hands-on sessions.
- Course papers and instructions will be collected in Canvas

Hands-on sessions: what we expect

Do not just Execute

What am I doing?

Why does it matter?

How do I interpret it?

- No advanced coding experience is expected.
- Code is provided; focus on the purpose and interpretation.
- Work together, but make sure you understand your own results.
- Errors are part of computational work, not evidence that you are bad at it.

Learn the lingo!

- Knowing the lingo is important
 - Counts, TPM, Metadata, Design, PCA, Contrasts, FDR, Enrichment etc etc
- Knowing them will make your life easier (i.e. easier to google or use AI on something :D)

Use of AI tools

Helpful friend, but not a scientific authority

Good uses

- Explain code
- Understand errors
- Clarify concepts

Be careful

- Do not blindly trust code
- Check outputs
- Verify interpretation

Your responsibility

- Scientific reasoning
- Data interpretation
- Assignment conclusions

AI can help you write code. It cannot decide whether your biological conclusion is valid.

Course elements and assessment

Individual final assignment

One-page reflection:
how and why omics /
systems biology could
support your research

Group journal club

20 min presentation
+ 10 min Q&A
+ opponent role

Hands-on

No submission
needed, but please do
them 😊

For the journal club, do not simply summarize the paper, evaluate it.

Opening Question

- 2 mins, discuss it with your neighbor

You have unlimited money and can measure ONE omics layer in your project. Which one would you choose, and what biological question would it answer?

[illegible]

14